

File Systems

Dr. Modhuparna Manna

What is a file system?

A filesystem is the method and structure an operating system uses to store, organize, manage, and retrieve data on storage devices like hard drives, SSDs, USB drives, memory cards, and CDs/DVDs.

A filesystem may answer the following:

- How to store files?
- Where to store files?
- How to find already stored files?
- What metadata each file has?
- How to manage free and used space?

Why are we studying about file systems?

Instead of pointing and clicking to an existing tool, one needs thorough knowledge of filesystems necessary to create tools and do deep digital forensic investigation.

It is critical to understand about file systems because:

- File deletion behaviour varies by filesystem
- Metadata reveals user actions
- Filesystem logs show user activity
- File carving depends on allocation structure

Windows File Systems

File System	Full Form	Use Case	Journaling	Max File Size	Notes
FAT12	File Allocation Table (12 bit)	Floppy disks	No	Very small	Legacy
FAT16	File Allocation Table (16 bit)	Old systems	No	2 GB	Legacy
FAT32	File Allocation Table (32 bit)	USB/SD cards	No	4 GB	Common removable media
exFAT	Extended File Allocation Table	Flash/SDXC	No	Very large	Modern portable devices
NTFS	New Technology File System	Internal disks	Yes	Very large	Default Windows file system
ReFS	Resilient File System	Servers	Yes	Very large	High redundancy
CDFS	Compact Disk File Systems	CD-ROM	No	N/A	Read-only
UDF	Universal Disk File Systems	DVD	No	N/A	Optical media

FAT12

- Oldest Windows/DOS file system.
- Used in floppy disks, early DOS systems.
- Stores file allocation using a 12-bit table.
- Easy to carve files due to simple structure.
- Maintains minimal metadata.

FAT16

- Used in DOS, Windows 3.x, early Windows 95
- Supported up to 2 GB partitions
- No journaling maintained
- Limited timestamps – always stores last modified date & time; can store creation date/time and last access *date only*; no high-precision or time zone info.

FAT: Characteristics

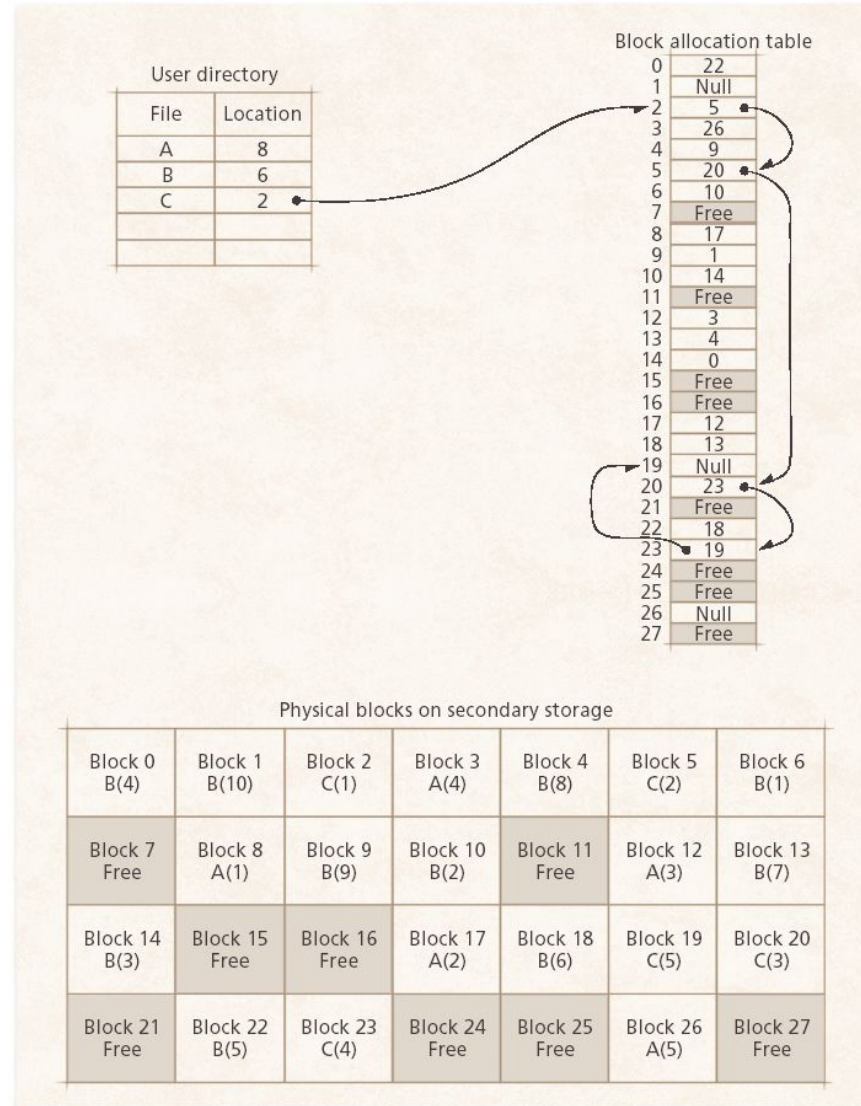
- FAT12/FAT16 (MS-DOS, Win95/98/NT/2000/XP)
 - Limited root directory size (512 entries)
 - No security features
 - 8.3 filenames
 - 12/16 bit cluster numbers
- FAT32 (Win95 w/ patches/98/2000/XP)
 - No limit on root directory size
 - No security features
 - Support for long filenames
- Basic difference is size of FAT entry—12, 16, or 32 bits.
- Affects maximum number of addressable clusters
- NT/2000/XP used NTFS by default; FAT12/16 only for compatibility.

FAT Timestamps

Limited time/date information for FAT files

- Last write date/time is always available
- Creation date/time is optional and may not be available
- Last access DATE ONLY is optional and may not be available
- FAT timestamp granularity: 2 seconds for time, 1 day for access date

FAT-based Filesystem: Schematic



FAT

- All integers are little endian
 - LSB first
- Four regions in FAT partition (in order)
 - Reserved Region
 - FAT Region
 - Root Directory Region (not for FAT32)
 - File and Directory Data Region
- Reserved region contains BPB
 - **B**IOS **P**arameter **B**lock
 - Contains config info for partition + bootstrap code
 - Bytes per sector, sectors per cluster, ...
 - Volume ID, # of FATs, ...
- FAT32 maintains a backup of the BPB at sector 6
- For FAT12/16, loss of sector 0 (storing BPB) can cause serious problems as there is no backup.

FAT

- The FAT (File Allocation Table) is simply a block of integers, one per cluster on the disk
- FAT12 is a little harder to deal with programmatically, because the FAT entries are 1.5 bytes (12 bits) long
- FAT16/FAT32 are easier, because FAT entries are either 2 bytes or 4 bytes long
- For FAT32, most significant 4 bits in FAT entries are not used

FAT: Root directory

- FAT12/FAT16: first sector of the root directory is relative to first sector of the FAT volume:
 - $\text{FirstRootDirSecNum} = \text{BPB_ResvdSecCnt} + (\text{BPB_NumFATs} * \text{BPB_FATSz16});$
- FAT32: root directory size is not limited and is a cluster chain, like other directories
- First cluster of root directory on FAT32 is stored in BPB_RootClus field
- Root dir does not have date or timestamps
- Does not have a filename
- Does not contain “.” or “..” entries

Root Directory Summary

- **Limited root directory size (FAT12/16):**
- Root directory has a fixed size (fixed number of entries)
- Cannot grow even if free space is available
- Limits how many files/folders can be created in the root
- Caused by the root directory being placed in a fixed region, not in cluster chains
- The fixed number of entries does not apply to the folders inside the root directory.

FAT: Directory Entry Format

FAT 32 Byte Directory Entry Structure

Name	Offset (byte)	Size (bytes)	Description
DIR_Name	0	11	Short name.
DIR_Attr	11	1	File attributes: ATTR_READ_ONLY 0x01 ATTR_HIDDEN 0x02 ATTR_SYSTEM 0x04 ATTR_VOLUME_ID 0x08 ATTR_DIRECTORY 0x10 ATTR_ARCHIVE 0x20 ATTR_LONG_NAME ATTR_READ_ONLY ATTR_HIDDEN ATTR_SYSTEM ATTR_VOLUME_ID The upper two bits of the attribute byte are reserved and should always be set to 0 when a file is created and never modified or looked at after that.
DIR_NTRes	12	1	Reserved for use by Windows NT. Set value to 0 when a file is created and never modify or look at it after that.
DIR_CrtTimeTenth	13	1	Millisecond stamp at file creation time. This field actually contains a count of tenths of a second. The granularity of the seconds part of DIR_CrtTime is 2 seconds so this field is a count of tenths of a second and its valid value range is 0-199 inclusive.
DIR_CrtTime	14	2	Time file was created.
DIR_CrtDate	16	2	Date file was created.
DIR_LstAccDate	18	2	Last access date. Note that there is no last access time, only a date. This is the date of last read or write. In the case of a write, this should be set to the same date as DIR_WrtDate.
DIR_FstClusHI	20	2	High word of this entry's first cluster number (always 0 for a FAT12 or FAT16 volume).
DIR_WrtTime	22	2	Time of last write. Note that file creation is considered a write.
DIR_WrtDate	24	2	Date of last write. Note that file creation is considered a write.
DIR_FstClusLO	26	2	Low word of this entry's first cluster number.
DIR_FileSize	28	4	32-bit DWORD holding this file's size in bytes.

FAT: Dir Entry Filenames

- Short filenames (8 + 3 characters)
- If first byte of name is 0xE5, then entry is deleted
- If first byte of name is 0x00, means no more entries, everything after this is **empty/free**
- Saves FAT driver resources, because other entries don't need to be checked
- If first byte is 0x05, then first byte represents 0xE5 (valid Japanese char)
- These characters are illegal:
 - Less than 0x20 (space), except 0x05 special case
 - 0x22, 0x2A, 0x2B, 0x2C, 0x2E, 0x2F, 0x3A, 0x3B, 0x3C, 0x3D, 0x3E, 0x3F, 0x5B, 0x5C, 0x5D, 0x7C
 - Generally, ASCII chars less than 0x20 are control characters, such as carriage return, line feed, bell, tab, backspace, etc.

Short Filename Storage

- `"foo.bar"` → `"FOO BAR"`
 - `"FOO.BAR"` → `"FOO BAR"`
 - `"Foo.Bar"` → `"FOO BAR"`
 - `"foo"` → `"FOO "`
 - `"foo."` → `"FOO "`
 - `"PICKLE.A"` → `"PICKLE A "`
 - `"prettybg.big"` → `"PRETTYBGBIG"`
-
- Note case is not significant
 - `"."` between primary filename and extension is implied (not actually stored)
 - Further, everything is space-padded

FAT: Dir Entry Details (2)

- Date format:
 - Bits 0–4: Day of month, valid value range 1-31 inclusive.
 - Bits 5–8: Month of year, 1 = January, valid value range 1–12 inclusive.
 - Bits 9–15: Count of years from 1980, valid value range 0–127 inclusive (1980–2107).
- Time Format:
 - A FAT directory entry time stamp is a 16-bit field that has a granularity of 2 seconds
 - Bits 0–4: 2-second count, valid value range 0–29 inclusive (0 – 58 seconds).
 - Bits 5–10: Minutes, valid value range 0–59 inclusive
 - Bits 11–15: Hours, valid value range 0–23 inclusive

FAT: Long Filename Support

- Long filename support is a hack, designed to be backwards compatible
- In particular, don't want to change on-disk structures in a way that breaks older FAT drivers, filesystem utilities, etc.
- Thus, long filenames are stored as a combination of a regular, short filename directory structure followed by a bunch of additional directory structures that will be “ignored” by non-compliant FAT versions

FAT: Long Filenames (2)

- Up to 255 characters in pathname component
- Total pathname no longer than 260 chars
- More supported characters
- Leading/trailing spaces ignored
- Internal spaces allowed
- Leading/embedded “.” allowed
- Trailing “.” are ignored

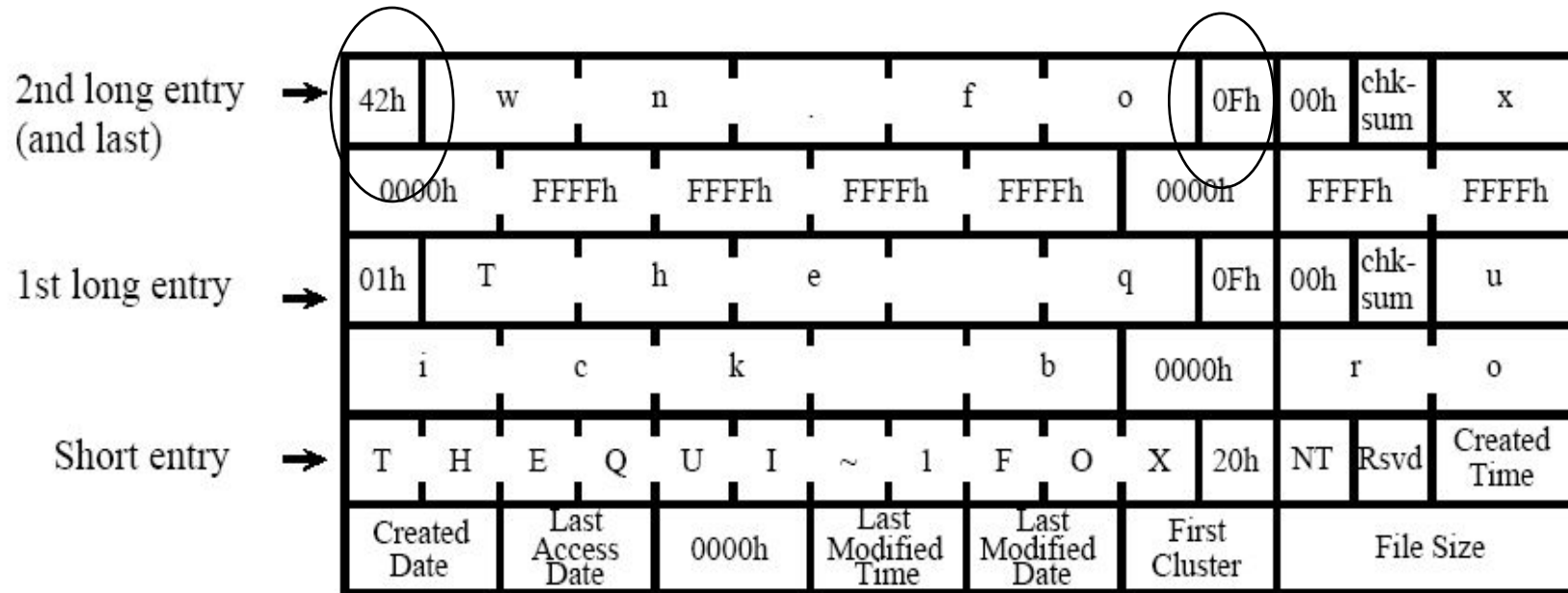
FAT: Long Filenames (3)

FAT Long Directory Entry Structure

Name	Offset (byte)	Size (bytes)	Description
LDIR_Ord	0	1	The order of this entry in the sequence of long dir entries associated with the short dir entry at the end of the long dir set. If masked with 0x40 (LAST_LONG_ENTRY), this indicates the entry is the last long dir entry in a set of long dir entries. All valid sets of long dir entries must begin with an entry having this mask.
LDIR_Name1	1	10	Characters 1-5 of the long-name sub-component in this dir entry.
LDIR_Attr	11	1	Attributes - must be ATTR_LONG_NAME
LDIR_Type	12	1	If zero, indicates a directory entry that is a sub-component of a long name. NOTE: Other values reserved for future extensions. Non-zero implies other dirent types.
LDIR_Chksum	13	1	Checksum of name in the short dir entry at the end of the long dir set.
LDIR_Name2	14	12	Characters 6-11 of the long-name sub-component in this dir entry.
LDIR_FstClusLO	26	2	Must be ZERO. This is an artifact of the FAT "first cluster" and must be zero for compatibility with existing disk utilities. It's meaningless in the context of a long dir entry.
LDIR_Name3	28	4	Characters 12-13 of the long-name sub-component in this dir entry.

ATTR_LONG_NAME = ATTR_READ_ONLY | ATTR_HIDDEN | ATTR_SYSTEM | **ATTR_VOLUME_ID**

FAT: Long Filenames



“the quick brown.fox”

Note that short name doesn't contribute to long name—long names preserve case, for one thing (tho case is not significant in matching filenames)

Long Name + rename

WinHex

File Edit Search Position View Tools Specialist Options Window Help

Drive E:

6 files, 0 dir.

Filename	Ext.	Size	Created	Modified	Accessed	Attr.
UTAustin2005forensics-series.ppt	ppt	13.4 MB	12/05/2005 13:36:...	12/05/2005 09:03:...	12/05/2005	A
Utaustin2005forensics.ppt	ppt	6.1 MB	10/18/2005 13:56:...	12/01/2005 12:43:...	01/29/2006	A
ifip2006.ppt	ppt	365 KB	01/26/2006 23:52:...	01/29/2006 21:40:...	01/30/2006	A
ifip2006.doc	doc	95.0 KB	01/26/2006 23:52:...	01/26/2006 12:18:...	01/29/2006	A
gumbo2.txt	txt	1.3 KB	04/30/2004 18:11:...	04/30/2004 18:11:...	04/30/2004	A
gumbo1.txt	txt	2.7 KB	04/30/2004 18:11:...	04/30/2004 18:11:...	04/30/2004	A
Free space		227 MB				

Offset

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F	Access	
0003E220	47	55	4D	42	4F	31	20	20	54	58	54	20	00	00	6A	91	9E	30	9E	30	00	00	6A	91	9E	30	02	00	FF	0A	00	00	GUMBO1 TXT	
0003E240	41	67	00	75	00	6D	00	62	00	6F	00	0F	00	96	32	00	2E	00	74	00	78	00	74	00	00	00	00	00	FF	FF	FF	FF	Aglaabla	
0003E260	47	55	4D	42	4F	32	20	20	54	58	54	20	00	00	6C	91	9E	30	9E	30	00	00	6C	91	9E	30	03	00	00	05	00	00	GUMBO2 TXT	
0003E280	42	66	00	6F	00	72	00	65	00	6E	00	0F	00	55	73	00	69	00	63	00	73	00	2E	00	70	00	00	00	70	00	74	00	Bfola	
0003E2A0	01	75	00	74	00	61	00	75	00	73	00	0F	00	55	74	00	69	00	6E	00	32	00	30	00	30	00	00	00	35	00	2D	00	la	
0003E2C0	55	54	41	55	53	54	7E	31	50	50	54	20	00	A8	03	6F	52	33	3D	34	00	00	63	65	81	33	04	00	00	1C	62	00	UTAUST*1PPT	
0003E2E0	43	69	00	65	00	73	00	2E	00	70	00	0F	00	75	70	00	74	00	00	00	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF	FF	FF	Citel
0003E300	02	66	00	6F	00	72	00	65	00	6E	00	0F	00	75	73	00	69	00	63	00	73	00	2D	00	73	00	00	00	65	00	72	00	la	
0003E320	01	55	00	54	00	41	00	75	00	73	00	0F	00	75	74	00	69	00	6E	00	32	00	30	00	30	00	00	00	35	00	2D	00	la	
0003E340	55	54	41	55	53	54	7E	32	50	50	54	20	00	95	97	6C	85	33	85	33	00	00	67	48	85	33	26	06	00	4E	D6	00	UTAUST*2PPT	
0003E360	49	46	49	50	32	30	30	36	50	50	54	20	18	9C	81	BE	3A	34	3E	34	00	00	13	AD	3D	34	8B	13	00	B4	05	00	IFIP2006PPT	
0003E380	49	46	49	50	32	30	30	36	44	4F	43	20	18	A7	81	BE	3A	34	3D	34	00	00	4D	62	3A	34	01	14	00	7C	01	00	IFIP2006DOC	
0003E3A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0003E3C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0003E3E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
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0003E420	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0003E440	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
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0003E4C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
0003E4E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
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0003E560	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
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0003E5E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00		
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Sector 497 of 506848

Offset: 3E340

= 85

Block: n/a

Size: n/a

Long Name + rename (2)

WinHex

File Edit Search Position View Tools Specialist Options Window Help

Drive E:

Drive E:

6+1 files, 0 dir.

Filename	Ext.	Size	Created	Modified	Accessed	Attr.
UTAustin2005-forensics-series.ppt	ppt	13.4 MB	12/05/2005 13:36...	12/05/2005 09:03...	02/09/2006	A
utAustin2005-forensics.ppt	ppt	6.1 MB	10/18/2005 13:56...	12/01/2005 12:43...	01/29/2006	A
Look at how long this directory entry is lo...	ppt	13.4 MB	12/05/2005 13:36...	12/05/2005 09:03...	02/09/2006	A
ifip2006.ppt	ppt	365 KB	01/26/2006 23:52...	01/29/2006 21:40...	01/30/2006	A
ifip2006.doc	doc	95.0 KB	01/26/2006 23:52...	01/26/2006 12:18...	01/29/2006	A
gumbo2.bt	bt	1.3 KB	04/30/2004 18:11...	04/30/2004 18:11...	04/30/2004	A
gumbo1.bt	bt	2.7 KB	04/30/2004 18:11...	04/30/2004 18:11...	04/30/2004	A
Free space		227 MB				

Offset

Offset	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F	10	11	12	13	14	15	16	17	18	19	1A	1B	1C	1D	1E	1F	Access	
0003E220	47	55	4D	42	4F	31	20	20	54	58	54	20	00	00	6A	91	9E	30	9E	30	00	00	6A	91	9E	30	02	00	FF	0A	00	00	GUMBO1 TXT	
0003E240	41	67	00	75	00	6D	00	62	00	6F	00	0F	00	96	32	00	2E	00	74	00	78	00	74	00	00	00	00	00	FF	FF	FF	FF	Agummblo	
0003E260	47	55	4D	42	4F	32	20	20	54	58	54	20	00	00	6C	91	9E	30	9E	30	00	00	6C	91	9E	30	03	00	0D	05	00	00	GUMBO2 TXT	
0003E280	42	66	00	6F	00	72	00	65	00	6E	00	0F	00	55	73	00	69	00	63	00	73	00	2E	00	70	00	00	00	70	00	74	00	Bfioirle	
0003E2A0	01	75	00	74	00	61	00	75	00	73	00	0F	00	55	74	00	69	00	6E	00	32	00	30	00	30	00	00	00	35	00	2D	00	utlata	
0003E2C0	55	54	41	55	53	54	7E	31	50	50	54	20	00	A8	03	6F	52	33	3D	34	00	00	63	65	81	33	04	00	00	1C	63	00	UTAUSTR	
0003E2E0	E5	69	00	65	00	73	00	2E	00	70	00	0F	00	75	70	00	74	00	00	00	FF	FF	FF	FF	FF	FF	00	00	FF	FF	FF	FF	atelsl	
0003E300	E5	66	00	6F	00	72	00	65	00	6E	00	0F	00	75	73	00	69	00	63	00	73	00	2D	00	73	00	00	00	65	00	72	00	afioirle	
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0003E340	E5	54	41	55	53	54	7E	32	50	50	54	20	00	95	97	6C	85	33	49	34	00	00	67	48	85	33	26	06	00	4E	D6	00	ATAUST	
0003E360	49	46	49	50	32	30	30	36	50	50	54	20	18	9C	81	BE	3A	34	3E	34	00	00	13	AD	3D	34	8B	13	00	B4	05	00	IFIP200	
0003E380	49	46	49	50	32	30	30	36	44	4F	43	20	18	A7	81	BE	3A	34	3D	34	00	00	4D	62	3A	34	01	14	00	7C	01	00	IFIP200	
0003E3A0	4B	73	00	65	00	72	00	69	00	65	00	0F	00	61	73	00	2E	00	70	00	70	00	74	00	00	00	00	00	FF	FF	FF	FF	Ksletr	
0003E3C0	0A	30	00	35	00	2D	00	66	00	6F	00	0F	00	61	72	00	65	00	6E	00	73	00	69	00	63	00	00	00	73	00	2D	00	0051-f	
0003E3E0	09	6E	00	67	00	20	00	55	00	54	00	0F	00	61	41	00	75	00	73	00	74	00	69	00	6E	00	00	00	32	00	30	00	inigt	
0003E400	08	6F	00	6F	00	6F	00	6F	00	6F	00	0F	00	61	6F	00	6F	00	6F	00	6F	00	6F	00	6F	00	00	00	6F	00	6F	00	00	lolo
0003E420	07	6C	00	6F	00	6E	00	67	00	20	00	0F	00	61	6C	00	6F	00	6F	00	6F	00	6F	00	6F	00	00	00	6F	00	6F	00	00	llo
0003E440	06	20	00	72	00	65	00	61	00	6C	00	0F	00	61	6C	00	79	00	20	00	6C	00	6F	00	6E	00	00	00	67	00	20	00	l	
0003E460	05	6F	00	6F	00	6B	00	20	00	77	00	0F	00	61	6F	00	77	00	20	00	69	00	74	00	20	00	00	00	69	00	73	00	lolo	
0003E480	04	73	00	20	00	6C	00	6F	00	6F	00	0F	00	61	6B	00	20	00	6C	00	6F	00	6F	00	6B	00	00	00	20	00	6C	00	l	
0003E4A0	03	63	00	74	00	6F	00	72	00	79	00	0F	00	61	20	00	65	00	6E	00	74	00	72	00	79	00	00	00	20	00	69	00	l	
0003E4C0	02	6F	00	6E	00	67	00	20	00	74	00	0F	00	61	68	00	69	00	73	00	20	00	64	00	69	00	00	00	72	00	65	00	l	
0003E4E0	01	4C	00	6F	00	6F	00	6B	00	20	00	0F	00	61	61	00	74	00	20	00	68	00	6F	00	77	00	00	00	20	00	6C	00	l	
0003E500	0C	4F	4F	4B	41	54	7E	31	50	50	54	20	00	95	97	6C	85	33	49	34	00	00	67	48	85	33	26	06	00	4E	D6	00	OOKAT	
0003E520	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	
0003E540	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	
0003E560	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	
0003E580	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	
0003E5A0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	
0003E5C0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	
0003E5E0	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	
0003E600	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	
0003E620	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	00	l	

Sector 498 of 506848

Offset: 3E500

24

= 76 Block:

n/a

Size: n/a

NTFS (New Technology File System)

NTFS Characteristics

- 64 bit cluster numbers
- ≈ 16 exabytes
- Security features
 - Can limit user access to files
 - Access Control Lists
 - Encryption
- Support for compression

NTFS (2)

- NTFS: Everything is a file
- Master File Table (MFT)
 - Table that describes files and directories
 - Location of first entry is stored in boot sector
 - First entry corresponds to MFT—MFT is also a file!
 - Each file and directory has an entry in MFT
 - Each MFT entry fixed size □ 1024 bytes long
 - (actual size of MFT entries stored in boot sector—currently, “always” 1024)
- First 42 bytes of MFT entry is header
- Remaining bytes store attributes

NTFS: More on MFT

- MFT starts small and is expanded as needed
- MFT entries are not removed once created
- Means: size of MFT increases if large numbers of files are created, but never shrinks

NTFS: More

- All data stored as attributes
- Resident attribute: inside MFT entry
- Non-resident attribute: stored on disk clusters outside MFT
- Many system files that contain filesystem metadata
 - \$LOGFILE: filesystem transaction file
 - \$VOLUME: serial number, creation time, ...
 - \$BITMAP: cluster allocation bitmap
 - \$BOOT: boot record for filesystem
 - \$BADCLUS: list of bad clusters for volume
 - ...others...
- In earlier versions of NTFS, these files are visible with, e.g., “dir /ah”
- Completely invisible in Win2000/XP/.../□

NTFS Files

- NTFS files are constructed of attributes
- \$FILE_NAME
 - Flags: read-only, hidden, system, encrypted, etc.
 - Ooops: modification, access, creation, MFT modification times...
- \$STANDARD_INFORMATION
 - “Official” modification, access, creation, MFT modification times
 - read-only, system, hidden flag (again)
 - encrypted / compressed / sparse flag
- \$DATA
 - File contents
 - For short files, may be stored in MFT entry
 - For longer files, non-resident and file data is stored in clusters outside the MFT
 - Support for sparse files
- Extensive creation/access/modification information for files, unlike FAT32
- DIR /Tx (where x = C, A, W) to see these times

Location of MFT in NTFS

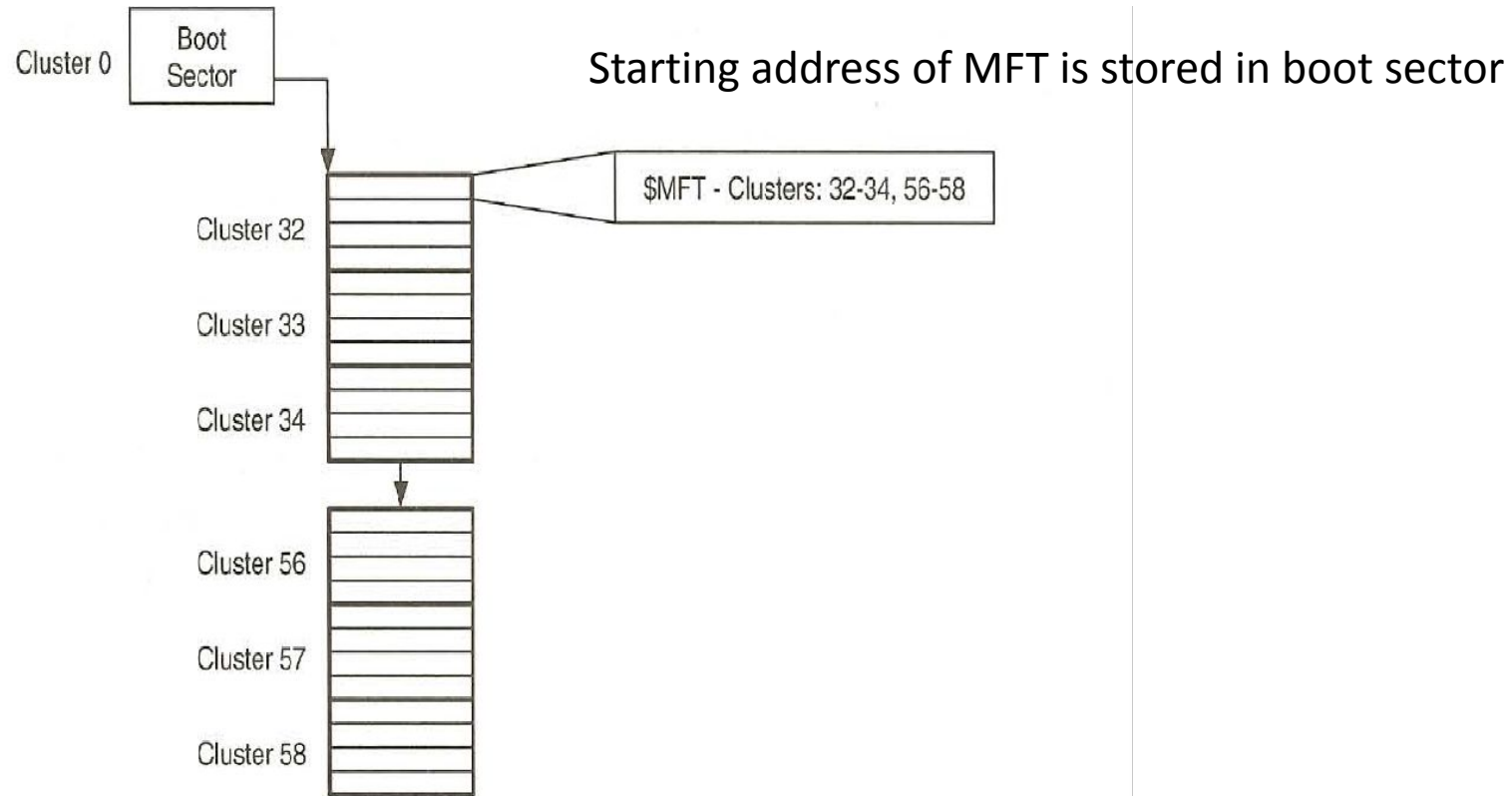


Figure 11.2 The relationship between the boot sector and \$MFT with respect to determining the layout of the MFT.

From: Filesystem Forensic Analysis (Carrier)

MFT Structure in NTFS

real size of MFT entry, deleted or not, file or directory, reuse time, etc.

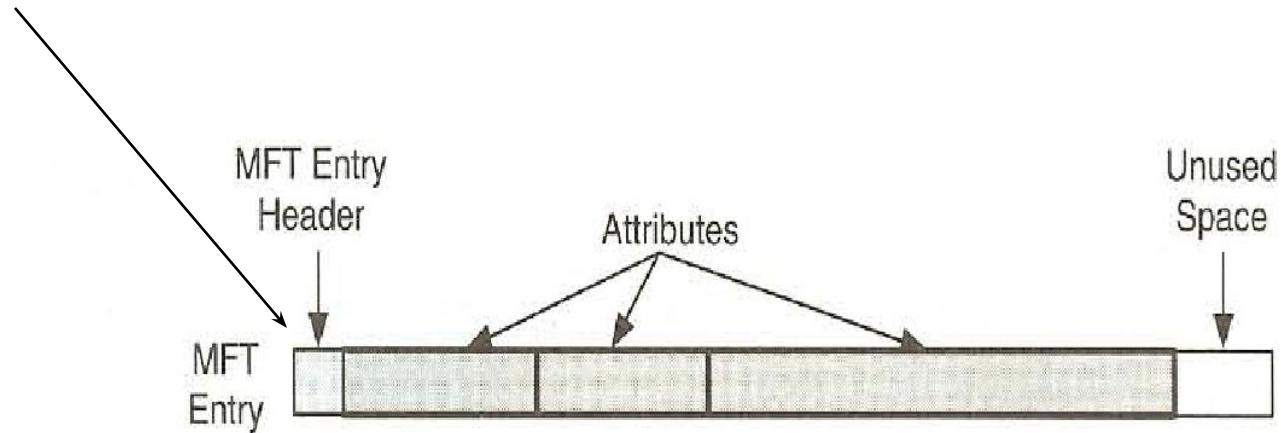


Figure 11.1 An MFT entry has a small header, and the rest of it is used to store different types of attributes. This entry has three attributes.

From: Filesystem Forensic Analysis (Carrier)

MFT Attribute Header: First 16 Bytes

Table 13.2 Data structure for the first 16 bytes of an attribute.

Byte Range	Description	Essential
0-3	Attribute type identifier	Yes
4-7	Length of attribute	Yes
8-8	Non-resident flag	Yes
9-9	Length of name	Yes
10-11	Offset to name	Yes
12-13	Flags	Yes
14-15	Attribute identifier	Yes

From: Filesystem Forensic Analysis (Carrier)

MFT: Resident Attribute Header

Table 13.3 Data structure for a resident attribute.

Byte Range	Description	Essential
0 - 15	General header (see Table 13.2)	Yes
16–19	Size of content	Yes
20–21	Offset to content	Yes

From: Filesystem Forensic Analysis (Carrier)

MFT: Non-resident Attribute

Table 13.4 Data structure for a non-resident attribute.

Byte Range	Description	Essential
0–15	General header (see Table 13.2)	Yes
16–23	Starting Virtual Cluster Number (VCN) of the runlist	Yes
24–31	Ending VCN of the runlist	Yes
32–33	Offset to the runlist	Yes
34–35	Compression unit size	Yes
36–39	Unused	No
40–47	Allocated size of attribute content	No
48–55	Actual size of attribute content	Yes
56–63	Initialized size of attribute content	No

From: Filesystem Forensic Analysis (Carrier)

Referencing Files in NTFS

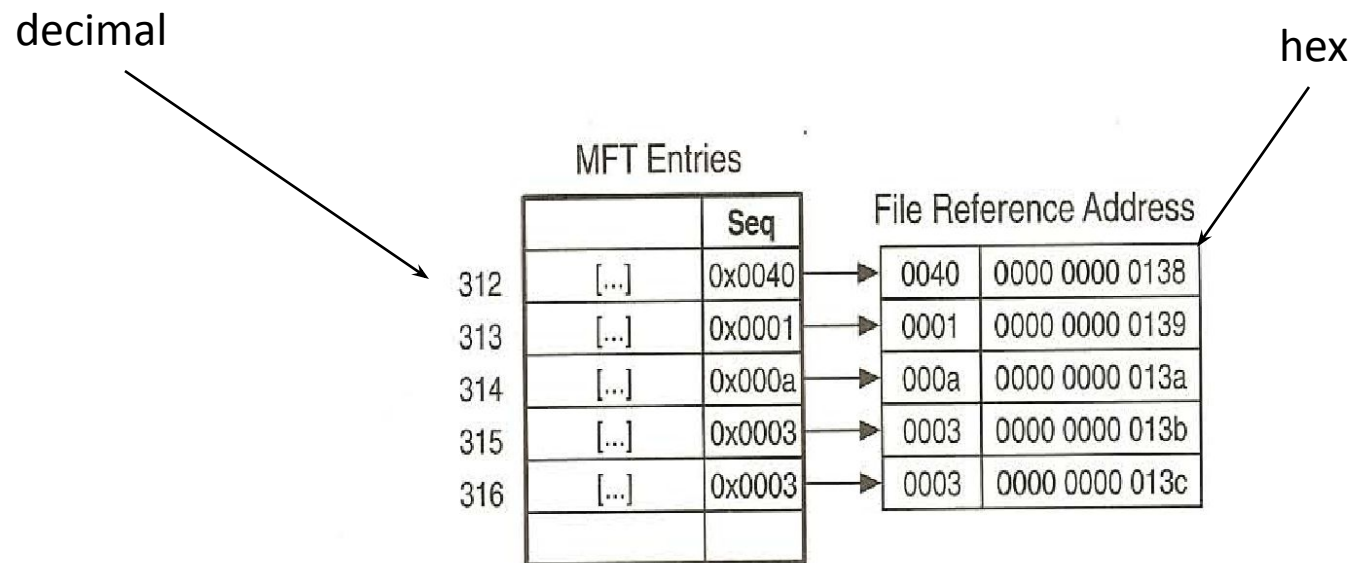


Figure 11.3 Example of the MFT entry address and sequence number combining to form a file reference address.

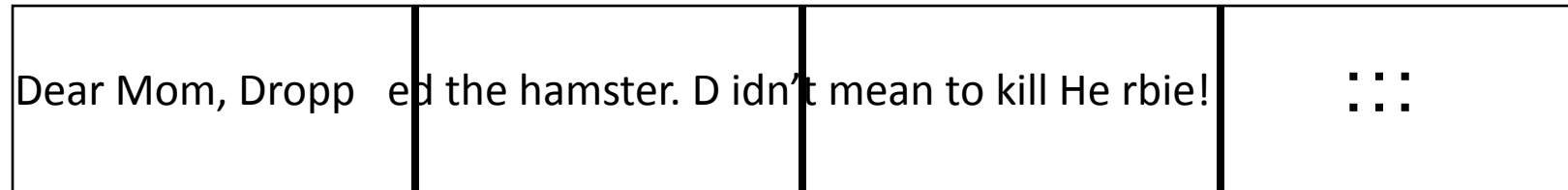
Files are referenced using 48 bit index into MFT + 16 bit sequence number.

Sequence # is updated when MFT entry is reused.

From: Filesystem Forensic Analysis (Carrier)

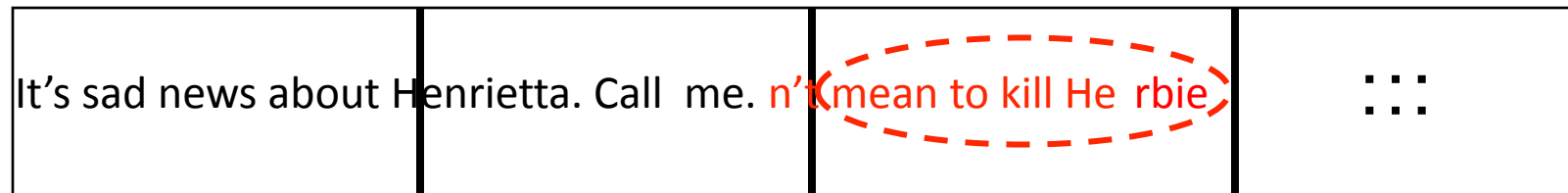
Slack Space (Simplified View)

MOM.TXT: "Dear Mom, Dropped the hamster. Didn't mean to kill Herbie!"



<< file is deleted and storage is reallocated >>

SORRY.TXT: "It's sad news about Henrietta. Call me."



file slack